
```

load('lin_sys.mat', 'C', 'sys');

Ts = 0.05; % Sampling time (in seconds)
sys_dt = c2d(sys, Ts, 'zoh'); % Discrete-time state-space model

Ad = sys_dt.A
Bd = sys_dt.B

% A matrix of augmented system
% the eye(3, 3) is tracking only the state of (x, y, z)
NA = [
    eye(3, 3), eye(3, 12);
    zeros(12, 3), Ad
]
NB = [zeros(3, 4); sys_dt.B]

CM = ctrb(NA, NB);
assert(rank(CM) == 15)

Q = diag([0.2, 0.2, 0.5, 2, 2, 20, 1, 1, 1, 2, 2, 0.1, 0.1, 0.1, 0.1])
R = eye(4)

[K, S, e] = dlqr(NA, NB, Q, R, zeros(15, 4))

Ki = K(:, 1:3)
Ks = K(:, 4:end)

pos_var = 2.5e-5
ang_var = 7.57e-5

qk_diag = repmat(1e-5, [1, 12])
qk_diag(6) = 2e-4
Qk = diag(qk_diag)

Rk = diag([repmat(pos_var, [1, 3]), repmat(ang_var, [1, 3])])

% Kalman filter
B1 = eye(12, 12)
[M, P] = dlqe(Ad, B1, C, Qk, Rk)
L = Ad * M

```

Ad =

Columns 1 through 7

1.0000	0	0	0.0494	0	0	0
0	1.0000	0	0	0.0494	0	-0.0122
0	0	1.0000	0	0	0.0494	0
0	0	0	0.9753	0	0	0

0	0	0	0	0.9753	0	-0.4844
0	0	0	0	0	0.9753	0
0	0	0	0	0	0	1.0000
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

Columns 8 through 12

0.0122	0	0	0.0002	0
0	0	-0.0002	0	0
0	0	0	0	0
0.4844	0	0	0.0122	0
0	0	-0.0122	0	0
0	0	0	0	0
0	0	0.0500	0	0
1.0000	0	0	0.0500	0
0	1.0000	0	0	0.0500
0	0	1.0000	0	0
0	0	0	1.0000	0
0	0	0	0	1.0000

Bd =

0	0.0000	0	-0.0000
-0.0000	0	0.0000	0
0.0001	0.0001	0.0001	0.0001
0	0.0003	0	-0.0003
-0.0003	0	0.0003	0
0.0030	0.0030	0.0030	0.0030
0.0019	0	-0.0019	0
0	0.0019	0	-0.0019
0.0001	-0.0001	0.0001	-0.0001
0.0750	0	-0.0750	0
0	0.0750	0	-0.0750
0.0050	-0.0050	0.0050	-0.0050

NA =

Columns 1 through 7

1.0000	0	0	1.0000	0	0	0
0	1.0000	0	0	1.0000	0	0
0	0	1.0000	0	0	1.0000	0
0	0	0	1.0000	0	0	0.0494
0	0	0	0	1.0000	0	0
0	0	0	0	0	1.0000	0
0	0	0	0	0	0	0.9753
0	0	0	0	0	0	0
0	0	0	0	0	0	0

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

Columns 8 through 14

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0.0122	0	0	0.0002
0.0494	0	-0.0122	0	0	-0.0002	0
0	0.0494	0	0	0	0	0
0	0	0	0.4844	0	0	0.0122
0.9753	0	-0.4844	0	0	-0.0122	0
0	0.9753	0	0	0	0	0
0	0	1.0000	0	0	0.0500	0
0	0	0	1.0000	0	0	0.0500
0	0	0	0	1.0000	0	0
0	0	0	0	0	1.0000	0
0	0	0	0	0	0	1.0000
0	0	0	0	0	0	0

Column 15

0
0
0
0
0
0
0
0
0
0
0
0
0.0500
0
0
1.0000

NB =

0	0	0	0
0	0	0	0
0	0	0	0
0	0.0000	0	-0.0000
-0.0000	0	0.0000	0
0.0001	0.0001	0.0001	0.0001
0	0.0003	0	-0.0003
-0.0003	0	0.0003	0

0.0030	0.0030	0.0030	0.0030
0.0019	0	-0.0019	0
0	0.0019	0	-0.0019
0.0001	-0.0001	0.0001	-0.0001
0.0750	0	-0.0750	0
0	0.0750	0	-0.0750
0.0050	-0.0050	0.0050	-0.0050

$Q =$

Columns 1 through 7

0.2000	0	0	0	0	0	0
0	0.2000	0	0	0	0	0
0	0	0.5000	0	0	0	0
0	0	0	2.0000	0	0	0
0	0	0	0	2.0000	0	0
0	0	0	0	0	20.0000	0
0	0	0	0	0	0	1.0000
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

Columns 8 through 14

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
1.0000	0	0	0	0	0	0
0	1.0000	0	0	0	0	0
0	0	2.0000	0	0	0	0
0	0	0	2.0000	0	0	0
0	0	0	0	0.1000	0	0
0	0	0	0	0	0.1000	0
0	0	0	0	0	0	0.1000
0	0	0	0	0	0	0

Column 15

0
0
0
0
0

```

0
0
0
0
0
0
0
0
0
0
0.1000

```

R =

```

1      0      0      0
0      1      0      0
0      0      1      0
0      0      0      1

```

K =

Columns 1 through 7

```

0.0000  -0.2520  0.3364  0.0000  -6.2934  12.0470  0.0000
0.2520  -0.0000  0.3364  6.2934  -0.0000  12.0470  3.1129
-0.0000  0.2520  0.3364  -0.0000  6.2934  12.0470  -0.0000
-0.2520  0.0000  0.3364  -6.2934  -0.0000  12.0470  -3.1129

```

Columns 8 through 14

```

-3.1129  8.0631  11.6872  -0.0000  0.1567  2.7120  -0.0000
-0.0000  8.0631  0.0000  11.6872  -0.1567  0.0000  2.7120
3.1129  8.0631 -11.6872  -0.0000  0.1567 -2.7120  0.0000
-0.0000  8.0631  0.0000 -11.6872  -0.1567  0.0000 -2.7120

```

Column 15

```

0.8989
-0.8989
0.8989
-0.8989

```

S =

1.0e+03 *

Columns 1 through 7

```

0.0050  0.0000  -0.0000  0.0589  0.0000  -0.0000  0.0177
0.0000  0.0050  -0.0000  0.0000  0.0589  -0.0000  0.0000
-0.0000  -0.0000  0.0179  -0.0000  -0.0000  0.3016  -0.0000
0.0589  0.0000  -0.0000  1.0184  0.0000  -0.0000  0.3539

```

0.0000	0.0589	-0.0000	0.0000	1.0184	-0.0000	0.0000
-0.0000	-0.0000	0.3016	-0.0000	-0.0000	8.1404	-0.0000
0.0177	0.0000	-0.0000	0.3539	0.0000	-0.0000	0.1448
0.0000	0.0177	-0.0000	0.0000	0.3539	-0.0000	0.0000
-0.0000	-0.0000	0.1178	-0.0000	-0.0000	4.1603	-0.0000
-0.0000	-0.0385	0.0000	-0.0000	-0.8565	0.0000	-0.0000
0.0385	0.0000	-0.0000	0.8565	0.0000	-0.0000	0.3809
0.0000	0.0000	-0.0000	0.0000	0.0000	0.0000	0.0000
-0.0000	-0.0043	0.0000	-0.0000	-0.1039	0.0000	-0.0000
0.0043	0.0000	-0.0000	0.1039	0.0000	-0.0000	0.0505
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Columns 8 through 14

0.0000	-0.0000	-0.0000	0.0385	0.0000	-0.0000	0.0043
0.0177	-0.0000	-0.0385	0.0000	0.0000	-0.0043	0.0000
-0.0000	0.1178	0.0000	-0.0000	-0.0000	0.0000	-0.0000
0.0000	-0.0000	-0.0000	0.8565	0.0000	-0.0000	0.1039
0.3539	-0.0000	-0.8565	0.0000	0.0000	-0.1039	0.0000
-0.0000	4.1603	0.0000	-0.0000	0.0000	0.0000	-0.0000
0.0000	-0.0000	-0.0000	0.3809	0.0000	-0.0000	0.0505
0.1448	-0.0000	-0.3809	0.0000	0.0000	-0.0505	0.0000
-0.0000	2.7559	0.0000	-0.0000	0.0000	0.0000	-0.0000
-0.3809	0.0000	1.1721	-0.0000	-0.0000	0.1836	-0.0000
0.0000	-0.0000	-0.0000	1.1721	0.0000	-0.0000	0.1836
0.0000	0.0000	-0.0000	0.0000	0.0115	-0.0000	0.0000
-0.0505	0.0000	0.1836	-0.0000	-0.0000	0.0406	-0.0000
0.0000	-0.0000	-0.0000	0.1836	0.0000	-0.0000	0.0406
0.0000	0.0000	-0.0000	0.0000	0.0316	-0.0000	0.0000

Column 15

0.0000
0.0000
0.0000
0.0000
0.0000
0.0000
0.0000
0.0000
0.0000
-0.0000
0.0000
0.0316
-0.0000
0.0000
0.1807

e =

0.9910 + 0.0087i
0.9910 - 0.0087i
0.9672 + 0.0484i

```

0.9672 - 0.0484i
0.9341 + 0.1273i
0.9341 - 0.1273i
0.9341 + 0.1273i
0.9341 - 0.1273i
0.9417 + 0.0000i
0.8971 + 0.0733i
0.8971 - 0.0733i
0.8971 + 0.0733i
0.8971 - 0.0733i
0.8604 + 0.0000i
0.8604 + 0.0000i

```

Ki =

```

0.0000 -0.2520 0.3364
0.2520 -0.0000 0.3364
-0.0000 0.2520 0.3364
-0.2520 0.0000 0.3364

```

Ks =

Columns 1 through 7

```

0.0000 -6.2934 12.0470 0.0000 -3.1129 8.0631 11.6872
6.2934 -0.0000 12.0470 3.1129 -0.0000 8.0631 0.0000
-0.0000 6.2934 12.0470 -0.0000 3.1129 8.0631 -11.6872
-6.2934 -0.0000 12.0470 -3.1129 -0.0000 8.0631 0.0000

```

Columns 8 through 12

```

-0.0000 0.1567 2.7120 -0.0000 0.8989
11.6872 -0.1567 0.0000 2.7120 -0.8989
-0.0000 0.1567 -2.7120 0.0000 0.8989
-11.6872 -0.1567 0.0000 -2.7120 -0.8989

```

pos_var =

```

2.5000e-05

```

ang_var =

```

7.5700e-05

```

qk_diag =

```

1.0e-05 *

```

Columns 1 through 7

```

    1.0000    1.0000    1.0000    1.0000    1.0000    1.0000    1.0000
Columns 8 through 12
    1.0000    1.0000    1.0000    1.0000    1.0000

qk_diag =
    1.0e-03 *
Columns 1 through 7
    0.0100    0.0100    0.0100    0.0100    0.0100    0.2000    0.0100
Columns 8 through 12
    0.0100    0.0100    0.0100    0.0100    0.0100

Qk =
    1.0e-03 *
Columns 1 through 7
    0.0100         0         0         0         0         0         0
         0    0.0100         0         0         0         0         0
         0         0    0.0100         0         0         0         0
         0         0         0    0.0100         0         0         0
         0         0         0         0    0.0100         0         0
         0         0         0         0         0    0.2000         0
         0         0         0         0         0         0    0.0100
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
Columns 8 through 12
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
    0.0100         0         0         0         0
         0    0.0100         0         0         0
         0         0    0.0100         0         0
         0         0         0    0.0100         0
         0         0         0         0    0.0100
         0         0         0         0    0.0100

```

Rk =

1.0e-04 *

0.2500	0	0	0	0	0
0	0.2500	0	0	0	0
0	0	0.2500	0	0	0
0	0	0	0.7570	0	0
0	0	0	0	0.7570	0
0	0	0	0	0	0.7570

B1 =

1	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	0	0	0	0
0	0	0	0	0	1	0	0	0	0	0	0
0	0	0	0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	0	0	1	0	0
0	0	0	0	0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0	0	0	0	1

M =

0.4929	0.0000	0.0000	0.0000	0.0115	0.0000
0.0000	0.4929	0.0000	-0.0115	-0.0000	-0.0000
0.0000	0.0000	0.5481	0.0000	-0.0000	-0.0000
0.5382	0.0000	-0.0000	0.0000	0.3332	0.0000
0.0000	0.5382	0.0000	-0.3332	-0.0000	-0.0000
-0.0000	0.0000	1.6393	0.0000	0.0000	0.0000
0.0000	-0.0347	0.0000	0.3338	-0.0000	0.0000
0.0347	-0.0000	-0.0000	-0.0000	0.3338	0.0000
0.0000	-0.0000	-0.0000	0.0000	0.0000	0.3354
0.0000	-0.0444	0.0000	0.2940	0.0000	0.0000
0.0444	-0.0000	-0.0000	-0.0000	0.2940	0.0000
-0.0000	-0.0000	-0.0000	0.0000	0.0000	0.2963

P =

Columns 1 through 7

0.0000	0.0000	0.0000	0.0000	0.0000	-0.0000	0.0000
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0000
0.0000	0.0000	0.0000	-0.0000	0.0000	0.0001	0.0000
0.0000	0.0000	-0.0000	0.0003	0.0000	-0.0000	0.0000

0.0000	0.0000	0.0000	0.0000	0.0003	-0.0000	-0.0000
-0.0000	0.0000	0.0001	-0.0000	-0.0000	0.0012	0.0000
0.0000	-0.0000	0.0000	0.0000	-0.0000	0.0000	0.0000
0.0000	-0.0000	-0.0000	0.0000	-0.0000	-0.0000	-0.0000
0.0000	-0.0000	-0.0000	0.0000	-0.0000	0.0000	0.0000
0.0000	-0.0000	0.0000	0.0000	-0.0000	0.0000	0.0000
0.0000	-0.0000	-0.0000	0.0000	-0.0000	-0.0000	-0.0000
-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	0.0000	0.0000

Columns 8 through 12

0.0000	0.0000	0.0000	0.0000	-0.0000
-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
-0.0000	-0.0000	0.0000	-0.0000	-0.0000
0.0000	0.0000	0.0000	0.0000	-0.0000
-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
-0.0000	0.0000	0.0000	-0.0000	0.0000
-0.0000	0.0000	0.0000	-0.0000	0.0000
0.0000	0.0000	0.0000	0.0000	0.0000
0.0000	0.0000	0.0000	0.0000	0.0000
0.0000	0.0000	0.0002	-0.0000	0.0000
0.0000	0.0000	-0.0000	0.0002	-0.0000
0.0000	0.0000	0.0000	-0.0000	0.0002

$L =$

0.5199	0.0000	-0.0000	0.0000	0.0320	0.0000
0.0000	0.5199	0.0000	-0.0320	-0.0000	-0.0000
-0.0000	0.0000	0.6290	0.0000	-0.0000	-0.0000
0.5423	0.0000	-0.0000	0.0000	0.4903	0.0000
0.0000	0.5423	-0.0000	-0.4903	-0.0000	-0.0000
-0.0000	0.0000	1.5988	0.0000	0.0000	0.0000
0.0000	-0.0370	0.0000	0.3485	-0.0000	0.0000
0.0370	-0.0000	-0.0000	-0.0000	0.3485	0.0000
0.0000	-0.0000	-0.0000	0.0000	0.0000	0.3502
0.0000	-0.0444	0.0000	0.2940	0.0000	0.0000
0.0444	-0.0000	-0.0000	-0.0000	0.2940	0.0000
-0.0000	-0.0000	-0.0000	0.0000	0.0000	0.2963

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